Jiayi (Olivia) Wang

Ph.D. Student, Computer Science & Engineering · University of South Florida
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RESEARCH INTERESTS

Multimodal affective computing · LLM multimodal reasoning · behavior understanding · AI for health & aging

EDUCATION

Ph.D., Computer Science & Engineering (GPA 3.97/4.0) Aug 2021 – Dec 2027 (expected)
University of South Florida, Tampa, FL · Advisor: Prof. Yu Sun · Robot Perception and Action Lab (RPAL)
Affiliated with the USF Center for Innovation, Technology and Aging (CITA)

M.S., Computer Science (GPA 3.96/4.0) May 2019
University of South Florida, Tampa, FL

B.E., Software Engineering Jun 2016
Inner Mongolia University, Hohhot, China

PUBLICATIONS

[1] **J. Wang**, G. Guo, A. Tabachnick, J. Su, H. Meng, S. Canavan, J. M. Templeton, D. Goldgof, and Y. Sun, "A Multimodal Dataset for Music-Induced Emotion and Micro-Behavior Analysis in Older Adults," *IEEE Transactions on Affective Computing*, accepted, 2026.

[2] J. Hausmann, **J. Wang**, M. Kneusel, S. Prescott, P. R. Mouton, Y. Sun, and D. Goldgof, "Automated Deep Learning Approach for Post-Operative Neonatal Pain Detection and Prediction Through Physiological Signals," in *Proc. IEEE 38th Int. Symp. Computer-Based Medical Systems (CBMS)*, pp. 801–806, 2025.

RESEARCH EXPERIENCE

AI-Assisted Music Intervention for Older Adults Jan 2023 – Present
Graduate Research Assistant, RPAL & CITA, University of South Florida

- Led construction of a therapist-annotated multimodal dataset of older adults (60+) during music listening: 17 participants, 200 clips (≈183K frames), 1,326 labeled micro-behavior instances.
- Designed the IRB-approved protocol and self-report instruments (valence, arousal, liking, familiarity); collaborated with a board-certified music therapist and gerontology researchers.
- Built a 3-stage annotation pipeline (therapist observation → LLM-structured extraction → manual validation) and a 12-class micro-behavior taxonomy across face, gaze, head, and body.
- Extracted de-identified features (facial AUs, gaze, head pose, upper-body keypoints) with OpenFace and OpenPose.
- Proposed Chain-of-Modality (CoM) prompting with GPT-4o for zero-shot micro-behavior detection: R@1@0.3 = 70.0%, mIoU = 0.524; mood-change prediction reached 0.88 accuracy from therapist-annotated behavior sequences.
- Proposed zero-shot KNN warm-up for personalized modality selection: balanced accuracy 0.770, closing 51% of the baseline–oracle gap (ongoing).
- Evaluated wearables (Fitbit Charge 6, Empatica EmbracePlus, Polar H10) for HRV/EDA capture.
- Funding: USF CREATE Award; Florida High-Tech Corridor Early-Stage Innovation Fund.

Multimodal Neonatal Pain Assessment Jan 2023 – May 2024
Engineering Team, RPAL Neonatal Pain Assessment Project, with USF Health · project page

- Processed multimodal NICU data (facial expression, body movement, heart rate, respiration rate, SpO2).
- Developed an LSTM pain classifier (74% AUC, 67.59% mAP, 5-fold CV); benchmarked CNN+LSTM, ViViT, and Transformer models for video-based pain detection.

AI-Based Stridor Detection with Bridge2AI-Voice Data Jun 2024 – Aug 2024
ML Modeling & Technical Support, Bridge2AI-Voice Program (NIH Common Fund)

- Built an end-to-end audio ML pipeline for stridor screening: 130 engineered acoustic features, PCA and feature selection, Random Forest and XGBoost; up to 93.2% accuracy.
- Team work received Honorable Mention at the Bridge2AI-Voice Hackathon and the 2nd Place Steven Dean Gray Award (ABEA, 2025).

GPU-Accelerated Index Search for Database Systems Aug 2021 – Dec 2022

University of South Florida

- Built a GPU-accelerated B+ tree in CUDA for parallel equality and range search, with a custom GPU memory allocator for dynamic index structures.

POSTERS & PRESENTATIONS

Apr 2026	"AI-Assisted Music Intervention for Older Adults via Multimodal Behavior Analysis," *USF CITA Open House*, poster.
Feb 2025	"AI-Based Precision Music Intervention for Alzheimer's Disease and Related Dementias (AD/ADRD)," *USF Center for Innovation, Technology and Aging*, talk.
Oct 2024	"AI-Based Precision Music Intervention for Alzheimer's Disease and Related Dementias (AD/ADRD)," *USF AI+X Symposium*, poster.
2023	"Precision Music Intervention through Computer Vision-Based Human Affect Assessment," *USF AI+X Symposium*, poster.

HONORS & AWARDS

2nd Place, Steven Dean Gray Award, American Broncho-Esophagological Association (team manuscript)	2025
Bridge2AI-Voice Scholar, NIH Common Fund Bridge2AI-Voice training program	2024
Honorable Mention, Bridge2AI-Voice Hackathon (9 teams, 4 universities)	2024
Grace Hopper Celebration Scholarship, University of South Florida	2022
University Graduate Fellowship, University of South Florida	2021
Outstanding Graduation Award, Inner Mongolia University	2016

TEACHING EXPERIENCE (TEACHING ASSISTANT, USF)

CAI 5005 Introduction to AI	Fall 2024
CIS 3213 Foundations of Cybersecurity	Fall 2023
COP 4703 Advanced Database for IT	Fall 2022, Spring 2023
COP 4538 Data Structures	Fall 2022
COP 4520 / COP 6527 Computing Massive Parallel Systems	Fall 2021
COP 6712 Database Management Systems	Fall 2021

MENTORING

- Mentored an undergraduate research volunteer from the University of Washington on the older-adult dataset project.
- Mentored USF CS students on wearable sensing: time-synchronized EDA and HR/HRV collection demos with Polar H10, Google Fitbit, and Empatica EmbracePlus, plus a device comparison table.
- Coordinate with behavioral-science students who run participant data collection; lead data analysis and modeling.

PROFESSIONAL SERVICE

Student Volunteer, IEEE International Conference on Robotics and Automation (ICRA) 2025

INDUSTRY EXPERIENCE

Quality Assurance, WebstaurantStore, Inc., Tampa, FL Mar 2020 – Jul 2021

- White/grey/black-box web testing in Agile teams; SQL-based database testing; cross-team root-cause triage.

Software Engineer, Greedy Cat, Inc., Tampa, FL Sep 2019 – Mar 2020

- Coordinated iOS/Android user testing and feedback collection; debugged and benchmarked the app against peer apps.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, C#, SQL, CUDA
ML / AI: PyTorch, scikit-learn, CNN/LSTM/Transformer, GPT-4o API, LLM prompting, computer vision, time series
Behavior & Health Sensing: OpenFace, OpenPose, facial AU/gaze/pose analysis, HRV/EDA/SpO2 signals, wearables
Research Methods: IRB protocol design, human-subject studies with older adults, expert annotation pipelines
Certifications: Oracle Database SQL Certified Expert; Oracle Database 11g Administrator Certified Associate